Machine Learning Research Digest

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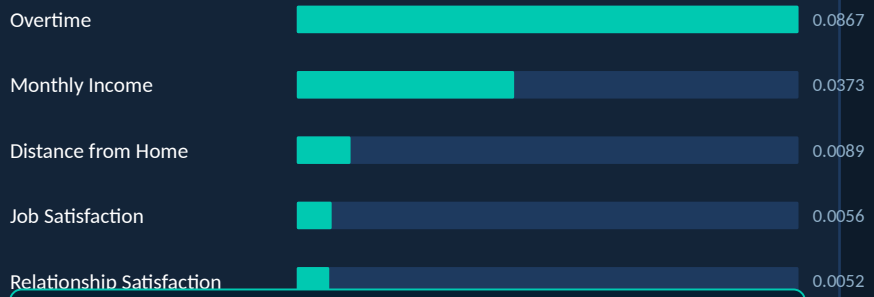
⬡ Explainable AI — Employee Attrition

IBM HR Dataset · XGBoost · SHAP · LIME · PDP · ELI5

MODEL PERFORMANCE

XGBoost	**85.91%**	79.2 ROC-AUC
Random Forest	85.81%	—
KNN	84.55%	—
SVM	83.87%	—
Decision Tree	80.66%	—

TOP ATTRITION DRIVERS



Insight: Reduce overtime, raise pay, and improve satisfaction before expanding headcount.

⬡ ML Credit Scoring — Airtime Loans

3M+ Loans · 41,391 Customers · Random Forest · Profitability Analysis

DATASET AT A GLANCE

3,002,667	41,391	72.5	$35.05
Loans Analysed	Customers	Avg Loans/Customer	Avg Borrowed

MODEL COMPARISON

Random Forest	**Acc: 82.3%**	Spec: 84.7%
Logistic Regression	Acc: 80.0%	Spec: 78.8%
Decision Tree	Acc: 79.7%	Spec: 79.8%

PROFITABILITY IMPACT

Without ML
Break-even Default Rate
8%

4×

With Random Forest
Break-even Default Rate
32%

Insight: Activate ML credit scoring dynamically above 2% default rate to maximize portfolio profitability.